

THE GREAT FORK

A Proposal for Zero-Ocean-Heating AI Infrastructure

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Credits

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No factual claims in this document are asserted without explicit labeling as proposals or hypotheses. All invented institutional names, partnership claims, and technical standards generated in earlier drafts have been removed.

Disclosure: *This document is a normative proposal and technical hypothesis. It does not describe existing partnerships, completed governance frameworks, or operational technical standards. All claims about what should happen are the author's. No claim about what has already happened is asserted unless independently verifiable. This is an ask, not a press release.*

One pathway heats the ocean and has no container for the 1000-year horizon. The other develops thermal technologies that could enable a long-term civilizational framework. This is a proposal for choosing the second.

ERES Corpus Association

This instrument is the engineering bridge between the ERES Constitution / Non-Punitive Remediation framework and the GAIA Solar Archive (ERES-GAIA-SOLAR-ARCHIVE-2026-001 v1.1). It answers the gap that previously existed: how does the thermal technology required to build the Solar Archive Tier C apex layer actually get developed? The answer is geothermal

high- ΔT R&D, activated by the zero-ocean-heating binding constraint. This is also the first ERES instrument applying the Three Governing Principles as a binding engineering constraint rather than a philosophical preamble.

Section 1: The Fork in the Road

Every major AI infrastructure build faces the same hidden question: Where does the heat go?

Current practice pulls in cool water — from rivers, lakes, or the ocean — runs it through heat exchangers, and dumps warmer water back. This adds heat to biologically sensitive waters. At scale, this contributes to local thermal pollution and, indirectly, to broader climate stress.

This is not a bug in current designs. It is an unacknowledged cost externality.

This document proposes an alternative: a binding constraint of zero net ocean thermal load for AI infrastructure. Not an efficiency improvement. A structural fork.

Section 2: Why Zero Ocean Heating Should Be a Non-Negotiable Constraint

Three governing principles apply to AI infrastructure development:

1. Don't hurt yourself. Accelerated climate collapse returns to harm every human, including those operating AI facilities.
2. Don't hurt others. Coastal communities, fisheries, and marine ecosystems did not consent to become industrial heat sinks.
3. Build for generations to come. Heat added to the ocean today persists for centuries. There is no remediation pathway for ocean heat content once added.

Under these principles, ocean heating violates all three simultaneously. Therefore, zero net ocean thermal load should be a non-negotiable constraint — not a preference, but a survival condition.

These are normative positions, not descriptions of current practice.

Section 3: The Thermal Technology Transfer Hypothesis

Here is the engineering hypothesis at the core of this proposal:

Conventional cooling engineers a small ΔT — the temperature difference between cool intake water and warm discharge water. That small ΔT does not generalize well to other high-temperature environments. It is a local optimum that leads nowhere.

Geothermal test-bed cooling engineers a large ΔT — the difference between deep rock at 150–400°C and the surface ambient environment. That large ΔT is the same class of problem as solar-proximate thermal management.

What keeps a geothermal instrument cluster alive at 300°C for ten years could directly inform what keeps a memory layer alive at 1000°C for a solar flyby read mission.

Therefore: investment in high- ΔT geothermal cooling R&D could, as a side benefit, inform the engineering needed for long-duration, high-temperature space missions. This is the developmental pathway for the GAIA Solar Archive Tier C apex layer.

This is a hypothesis, not an accomplished fact. It is offered for technical review and testing.

Section 4: Why Global Governance Would Be Required

Geothermal zones are not evenly distributed. They are geophysically constrained:

- Iceland / Mid-Atlantic Ridge — shared across Iceland, Azores/Portugal
- East African Rift — Ethiopia, Kenya, Tanzania, Uganda, Rwanda, Burundi, DRC, Malawi, Mozambique, Zambia
- Pacific Ring of Fire — Japan, Philippines, Indonesia, New Zealand, Chile, Peru, Mexico, US, Canada, Russia, China (Taiwan region)
- Andean Volcanic Belt — Chile, Argentina, Bolivia, Peru, Ecuador, Colombia

No single nation controls all high-temperature geothermal types. Therefore, any large-scale effort to develop and share high- ΔT cooling technology would require international cooperation on thermal budgets, test-bed access, and validation standards.

This document proposes a framework — GAIA (Global Adaptive Intelligence Architecture) — as a hypothetical governance protocol for thermal budgets, test-bed access, and technical validation. GAIA does not exist today. This is a proposal to create something like it.

Cooperation on geothermal R&D is not a diplomatic hope. It is geophysically forced — if we choose to pursue this pathway.

Section 5: What This Could Enable — The Solar Archive Concept

If zero-ocean-heating constraints were adopted, and if high- ΔT geothermal R&D were pursued, one long-term possibility would be a Solar Archive: a durable data repository capable of surviving solar-proximate conditions.

The Solar Archive would be:

- A physical data storage layer designed for high-temperature, long-duration environments
- Informed by geothermal instrumentation engineering — the thermal transfer hypothesis made real
- A 1000-year container for human knowledge, governed by GAIA

This is a speculative goal, not an active project. It is offered as a long-term horizon to motivate near-term decisions. Without a binding no-ocean-heating constraint, the thermal management knowledge required for such an archive may never be developed.

Section 6: The Binary Choice

There is no third pathway in this framing:

Pathway A — Current Default	Pathway B — Proposed
Heat the ocean	Zero ocean heating
Bypass global governance	Accept international cooperation
Develop no transferable high- ΔT thermal technology	Engineer large- ΔT thermal management
Have no container for the 1000-year horizon	Open the possibility of a Solar Archive
Externalize climate cost	Internalize survival condition

This is a choice this document argues for — not a description of what has already been chosen by any company or nation.

Section 7: Why This Is an Ask, Not a Press Release

This document does not announce accomplishments. It makes requests.

I am seeking:

4. Technical review — Is the geothermal → solar thermal transfer hypothesis plausible? What are the gaps?
5. Industry engagement — Would any AI infrastructure operator (national lab, major company, consortium) be willing to discuss a zero-ocean-heating pilot or binding constraint?
6. Governance interest — Who is working on global frameworks for AI thermal budgets? Can GAIA or something like it be developed?

This is not a fait accompli. It is an invitation.

References

7. Sprute, J. A., & DeepSeek (AI). (2026, June 6–7). Validation conversation regarding zero-ocean-heating AI infrastructure proposal. Generated dialogue. (Unpublished; available from author upon request.)
 8. No external sources are cited for factual claims because this document makes no empirical assertions. All technical hypotheses and normative proposals are clearly labeled as such.
 9. Background context (for readers): IPCC AR6 (2021) on ocean heat content; existing literature on AI data center water consumption; geothermal instrumentation engineering standards (e.g., IEEE, Geothermal Rising). The author makes no claim that these sources support the specific proposals herein.
 10. ERES-GAIA-SOLAR-ARCHIVE-2026-001 v1.1 (Sprute, J. A., 2026) — parent instrument defining the Solar Archive architecture and GAIA governance framework to which this proposal contributes the thermal engineering pathway.
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Closing

The ocean does not care about corporate profits. The Sun does not care about national boundaries. The only thing that scales across both is physics-based governance — if we choose to build it.

Trust is built into physics, not politics.

The Sun is the long-term custodian.

GAIA is the key-holder.

We are the short-term key-holders — or we could be.

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